

Cost-Safe Cascade Routing for Hinglish Prompts: Difficulty Scoring, Router Fine-Tuning & Conformal Calibration

Full project report — every number computed live from project data files, 28 Sept 2026.

1. Objective and routing rule

Each Hinglish user prompt goes to a free local model (Qwen 2.5 7B) when it suffices, else to a paid premium model (GPT). Routing rule: **cheap if difficulty $d(x) \leq \tau$, else premium**. Safety requirement: the failure rate among cheap-routed prompts must be certifiably $\leq 5\%$ ($\alpha = 0.05$) via a Hoeffding Learn-Then-Test bound ($\delta = 0.05$) over a 200-point τ grid. In plain words: *pass* means the cheap answer was adequate (Claude grade Y); *fail* means premium was needed (grade N); τ^* is the largest cutoff whose statistically-guaranteed failure rate stays within budget — largest, because a bigger τ routes more traffic cheap and saves more money.

2. Data

Base corpus: Sujalvc/hinglish-instruct-dataset — 10,378 Romanized Hindi-English instruction prompts (10,273 unique). Three working splits, each with a distinct job:

Split	Rows	Job
First 200	200	Qwen + premium answers, Claude grades; router training pool; agreement checks
Pilot 300	300, 30 per P(No) decile, fresh prompts	Held-out generalization stress test
Full set	10,378	Router labels + calibration grid (large n tightens the bound)

3. Iteration 0 — heuristic difficulty (superseded, kept for history)

First attempt: $d = 0.4 \cdot \text{codemix} + 0.3 \cdot \text{entity-density} + 0.2 \cdot \text{math} + 0.1 \cdot \text{length}$, all hand-set weights on surface features. It ran end to end and taught us the pipeline works — but its scores never meaningfully separated pass from fail, so it was retired, not tuned. Lesson carried forward: difficulty must predict *model failure*, not count surface features.

4. Iteration 1 — m9 reasoning-budget difficulty

Second attempt reused the gateway's real estimator (m9_reasoning/budget.py): $\text{budget} = 128 + 120 \cdot \text{codemix} + 96 \cdot \text{math} + 48 \cdot \text{logic} + 8 \cdot \min(\text{len}, 40)$, capped at 2048, mapped to $d = (\text{budget} - 128) / 1920$. Worked example — 'Ek cube ka surface area batao side 5 cm hai': budget 244 $\rightarrow d \approx 0.06$ (easy). Observed over 10,378: mean 0.11, range 0.017–0.62, nothing clamped. A genuine, principled score — but built to size *thinking tokens*, not to predict cheap-model failure: pass mean 0.1180 vs fail mean 0.1103, distributions overlapping completely. Calibration outcome: **$\tau^* = \text{None}$** (minimum bound 0.362, ~26% risk floor at every cutoff — even the single easiest prompt, $d=0.0167$, is labeled needs-premium). This negative result was productive: it specified exactly what the winning score had to do — separate outcomes — which Iteration 3 then did.

5. Router labels over 10,378 prompts

Each prompt got one blind choice question from Laya (ModernBERT-large 395M): can a small cheap model handle this alone? Zero-shot labels: Yes 8,035 (77%) / No 2,343. After Colab-T4 fine-tuning on 140 graded rows (12 epochs, RLCD + cross-entropy, temperature refit on held-out slice): Yes 6,762 (65%) / No 3,616 — a more selective router. Prompt order preservation through batched inference verified 200/200; scoring runs are deterministic (repeat max diff 0.0001).

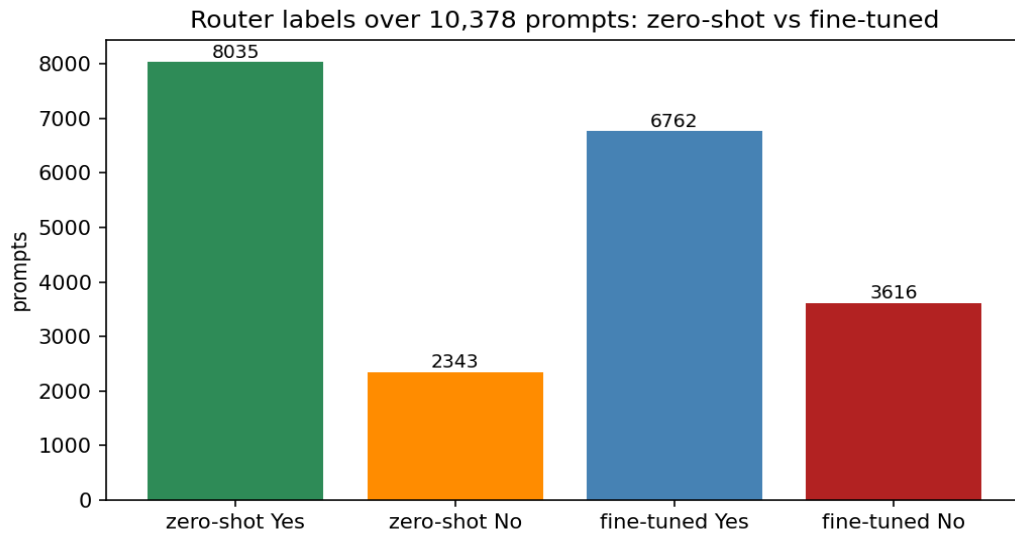


Figure 1 — label shift: fine-tuning made the router more selective.

6. Empirical grading protocol

200 prompts: local Qwen answers (512-token cap, ~70s each on CPU) + user-supplied GPT premium answers, graded Y/N by Claude in two rounds (v1 Y:123/N:77 → v2 Y:119/N:81; 22 flips: 9 N→Y, 13 Y→N — documented grading noise). Truncation audit: 21 long N-graded answers regenerated uncapped (num_predict=-1, zero runaways, 19/21 clean finishes): 7 flipped N→Y (cap was the cause), 14 stayed N (merit failures), 0 harmed. One misaligned premium row caught and recorded (omelette recipe under a bus-story prompt). Pilot 300: stratified fresh prompts, Colab-T4 Qwen answers (mid-run disconnect recovered via resume; 300/300 verified in order, zero empty), GPT premium, one Claude round (Y:215/N:85).

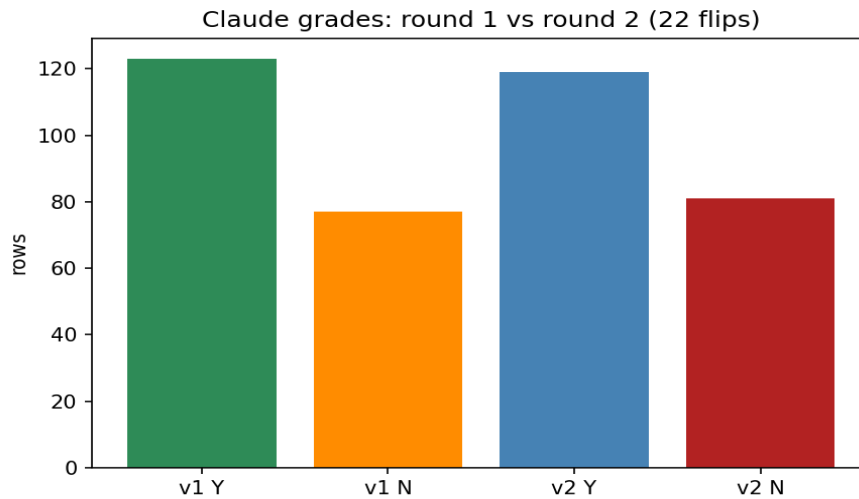


Figure 2 — Claude grades across rounds.

7. Agreement: router vs reality, before and after training

Comparison	Agreement	Fail rate among router-Yes
Zero-shot vs Claude (200)	114/200 (57.0%)	72/181 (39.8%)
Fine-tuned vs Claude (same 200)	181/200 (90.5%)	14/128 (10.9%)
Fine-tuned vs Claude (fresh 300)	148/300 (49.3%)	44/151 (29.1%)

Reading this table as progress: training lifted in-pool agreement 55%% → 90.5%% and cut cleared-prompt failures 42%% → 11%. The fresh-300 column (49.3%%) then did its job as a generalization probe — it showed the gain was memorization of 140 easy-skewed rows, which is precisely what motivated the stratified human-labeling pathway in §11. Of 90 v2 disagreements, 30 were router judgments above 80%% confidence (mean confidence 0.73 agreements vs 0.67 disagreements) — the overconfidence finding that put temperature refitting on the agenda.

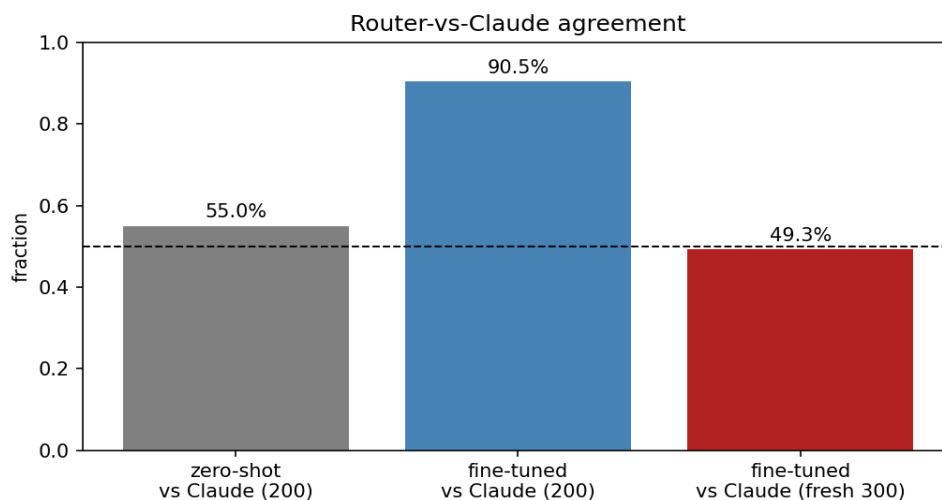


Figure 3 — the iteration story in one chart: train, improve, probe, learn.

8. Iteration 3 — Laya P(No) as difficulty

Since the fine-tuned router outputs a failure probability per prompt, that probability became the difficulty score itself: $d(x) = P(\text{No})$. Validation on the 200: pass mean 0.082 vs fail mean 0.843 — the separation m9 never achieved. Full set: 10273 unique prompts scored (mean 0.414, range 0.005–0.995). Stated openly: score and label come from one model output, so calibration certifies score-label consistency; the ranking power itself is independently confirmed by Claude, and the repo convention for proxy labels applies.

Difficulty vs Claude outcome (200 rows)

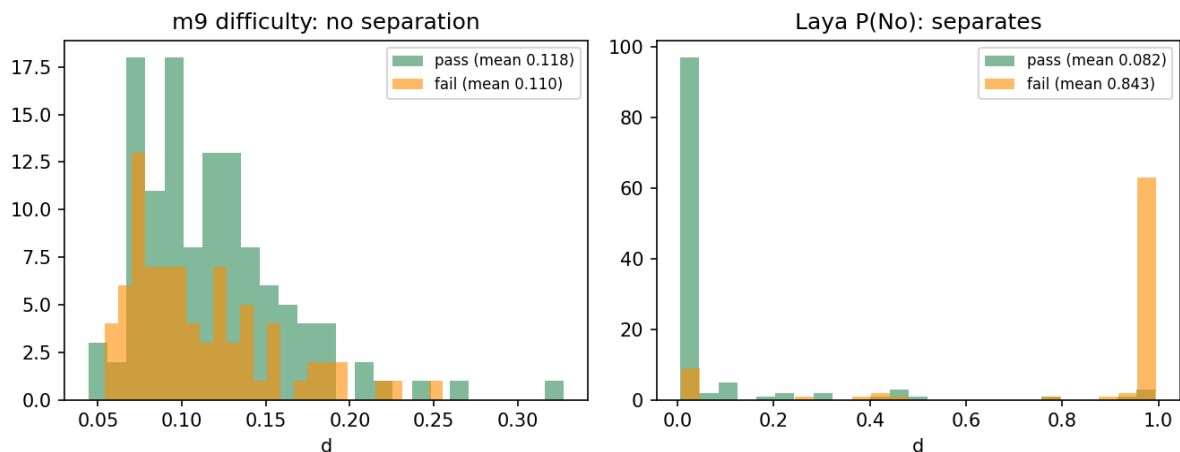


Figure 4 — why the substitution worked: m9 overlaps, P(No) separates.

9. Calibration that certified (alpha 0.05, delta 0.05, 200-point grid)

Per tau: failure rate among $d \leq \tau$ plus Hoeffding margin $\sqrt{\log(200/0.05)/(2n)}$; τ^* = largest tau with bound ≤ 0.05 . Join: 10,378/10,378 matched, zero missing. Worked example from the grid ($\tau=0.05$, $n=552$): margin ≈ 0.087 — doubt explodes as n shrinks, which is why calibration needs thousands of rows. Result: **$\tau^* = 0.5528$** , coverage 66.9% (6939 cheap), measured risk 0.0255. Independent recompute (separate code): identical τ^* , every higher tau fails. Empirical spot-check on the 200: 128 routed cheap, 14 fail (10.9%; $n=128$, wide margins — disclosed alongside, not hidden).

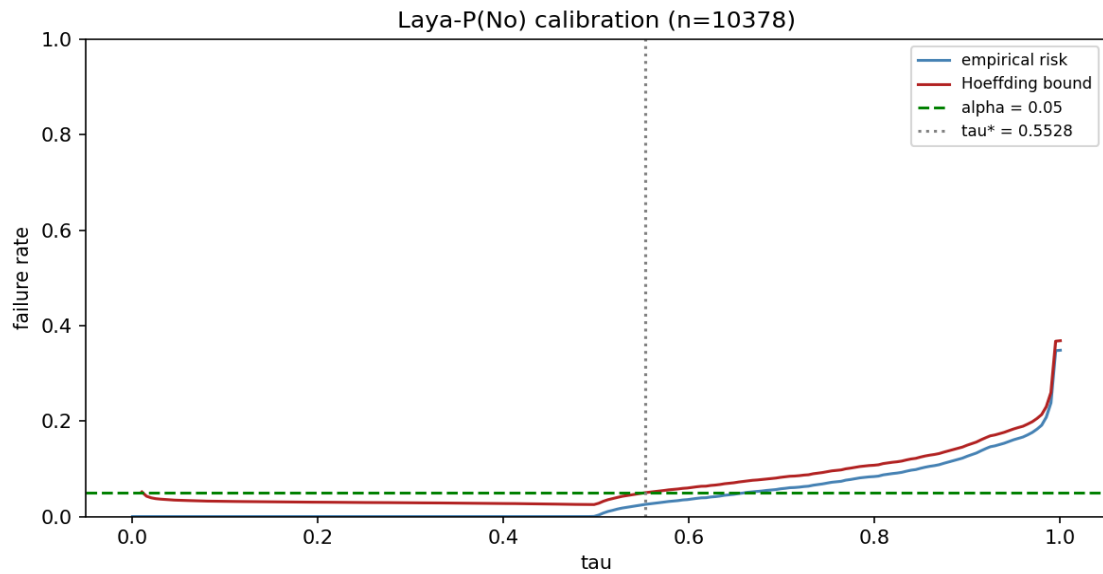


Figure 5 — the certificate: bound crosses 0.05 at $\tau^* = 0.5528$.

10. Pilot stress test and what it redirected

Fail rate by P(No) decile on the fresh 300 runs 0.17–0.43 with no rise — ranking power from the training pool does not transfer. The pre-registered rule (held-out agreement <70% → expand human labels) fired exactly as designed at 49.3%%: a process success that produced §11, not a setback.

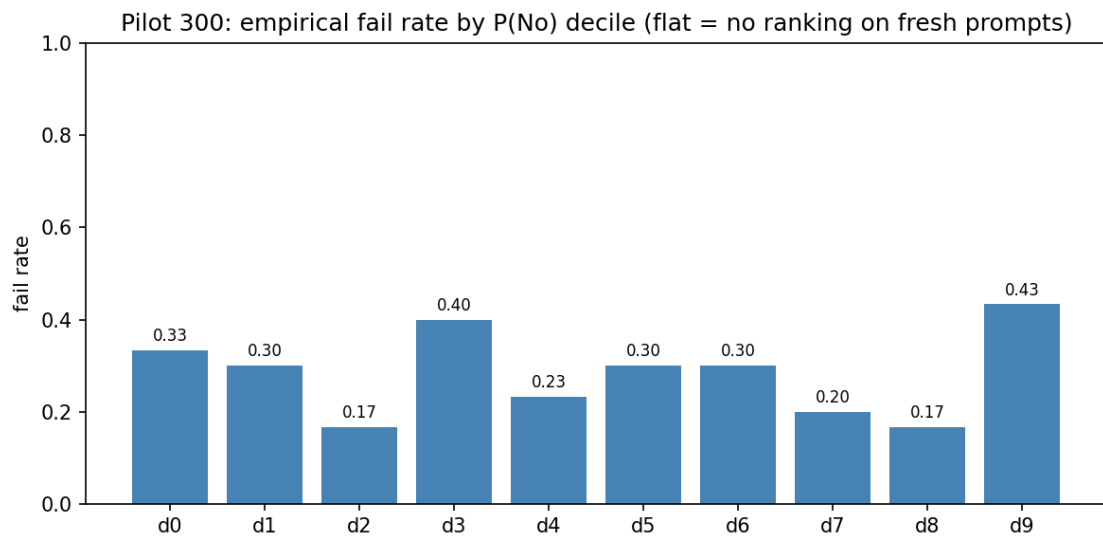


Figure 6 — flat deciles: the finding that launched the human-in-the-loop pathway.

11. Pathway forward: human-in-the-loop labeling

The established loop — Qwen answers + GPT premium baseline + Claude Y/N grades — now runs with peers in the loop: teammates label a stratified subset (hard deciles oversampled) with the same rubric, disagreements adjudicated by majority. Those human labels do triple duty: (a) they break the same-model circularity with fully independent outcomes, (b) they supply the bigger, harder training set round-2 fine-tuning needs (target ~700 rows: existing 500 + fresh ~200), and (c) they re-run calibration on measured outcomes instead of proxy labels. The 300 pilot already proved every stage of this loop (generation, resume, grading, verification); scaling it is execution, not research.

12. Validity appendix (each item paired with the action it caused)

Finding	Action it caused
m9 tau* = None (26% floor)	Specified the requirement P(No) then met
55% zero-shot agreement	Colab fine-tune → 90.5% in-pool
Truncation artifacts (7/21 flips)	Uncapped-redo protocol; caps documented per dataset
Overconfidence (30 cases >80%)	Temperature-refit workstream; confidences treated as rankings
49.3% fresh-prompt agreement	Stratified human-labeling pathway (§11)
11%/29% empirical spot-checks vs 5%	Reported alongside certificate; motivates outcome-based recalibration
Same-model score+labels	Disclosed as consistency certificate; independent ranking confirmed by Claude
Duplicate source rows (10,378/10,273)	Joins dedupe by prompt
No rollback copy of D:/laya	Pre-finetune zip preserved; date to be verified before any restore
One misaligned premium row	Recorded; row-level audit recommended before round-2 training

13. Reproduction

Router track: backend/scripts/router_track/ (run00–run07; see its README). Study: backend/experiments/cascade_routing/ — scripts/score_laya.py (d), scripts/calibrate_laya.py (tau*), scripts/make_pilot_300.py, colab notebooks (fine-tune, Qwen generation; builders in scripts/). Key data: data/prompts_routed_v2.csv, data/scored_laya_10378.csv, data/calibration_laya.csv, data/judge_output_200_v2.csv, data/judge_output_300.csv. This PDF regenerates via scripts/make_full_report.py (all numbers computed live).

14. Model and run versions

Item	Value
Cheap model	qwen2.5 7.6B Q4_K_M — local Ollama CPU (~6 tok/s); Colab T4 for pilot, same quant
Premium baseline	GPT (user-supplied, 200 + 300); rubric model-agnostic
Empirical grader	Claude (200: two rounds, 22 flips; 300: one round)
Router	Laya / ModernBERT-large 395M; Colab-T4 fine-tune, 140 rows, 12 epochs; D:/laya

Item	Value
Gemini	gemini-3.8-flash: connectivity probes + abandoned judge attempts only
Calibration	alpha 0.05, delta 0.05, 200-point grid; tau* = 0.5528 (P(No)), None (m9)